Chapter 12: Files and File Organization

Technical / interview questions, exercises, multiple choice questions and answer key for Chapter 12 (Files and File Organization) of Data Structures.

Technical / Interview Questions

1. Define a file in computer systems and explain its purpose.
2. What is file management in operating systems?
3. Explain the concept of data hierarchy with an example.
4. What is the difference between bit and byte?
5. Explain the relationship between field, record, and file.
6. What is a database and how does it relate to files?
7. Define file attributes and explain their significance.
8. List common file attributes maintained by operating systems.
9. What is the difference between logical file structure and physical file structure?
10. Explain the difference between text files and binary files.
11. What are the advantages of binary files over text files?
12. What are the disadvantages of binary files?
13. What are basic file operations supported by most operating systems?
14. Explain the operations open, read, write, and close.
15. What is the purpose of the seek operation in file processing?
16. What is file organization and why is it important?
17. Explain heap file organization.
18. Explain sequential file organization.
19. What are the advantages of sequential file organization?
20. What are the limitations of sequential file organization?
21. Explain indexed file organization.
22. What is indexed sequential file organization?
23. Explain the concept of an overflow area.
24. What is hash file organization?
25. Compare sequential and hashed file organization.
26. What is an index in file systems?
27. Explain the purpose of indexing.
28. What is a primary index?
29. What is a secondary index?
30. What is the difference between dense and sparse indexing?
31. Explain how indexing improves search efficiency.
32. What is a multi-level index?
33. Explain the concept of tree-based indexing.
34. What are the advantages of B-tree indexing?
35. Explain how file organization affects data retrieval performance.

Programming Exercises

1. Implement the following in C / C++ / Python / Java.

2. Write a program to create a text file and write user input into it.
3. Write a program to read and display the contents of a text file.
4. Write a program to append data to an existing file.
5. Write a program to copy contents from one file to another file.
6. Write a program to count the number of lines in a text file.
7. Write a program to count words in a file.
8. Write a program to count characters in a file.
9. Write a program to search for a word in a text file.
10. Write a program to replace a word in a text file.
11. Write a program to delete a specific record from a file.
12. Write a program to store student records in a file.
13. Write a program to retrieve student records from a file.
14. Write a program to update a record in a file.
15. Write a program to display all records from a file.
16. Write a program to sort records stored in a file.
17. Write a program to merge two files.
18. Write a program to split a large file into smaller files.
19. Write a program to store employee records in a binary file.
20. Write a program to read records from a binary file.
21. Write a program to update records in a binary file.
22. Write a program to search for a record in a binary file using roll number.
23. Write a program to demonstrate sequential file access.
24. Write a program that simulates indexed file access.
25. Write a program that demonstrates random file access.
26. Write a program to maintain a student database using files.
27. Write a program to calculate average marks from records stored in a file.
28. Write a program to remove duplicate records from a file.
29. Write a program to store and retrieve records using file pointers.
30. Write a program to demonstrate binary vs text file size comparison.
31. Write a program to simulate a simple index for file searching.

Multiple Choice Questions

1. The smallest unit of data is

(a) Byte
(b) Bit
(c) Record
(d) Field

2. A collection of related fields forms a

(a) File
(b) Record
(c) Byte
(d) Database

3. A collection of related records is called

(a) File
(b) Field
(c) Bit
(d) Byte

4. A group of files forms a

(a) Record
(b) Database
(c) Field
(d) Bit

5. Which of the following is a file attribute?

(a) File name
(b) File size
(c) File type
(d) All of the above

6. Which file type stores data in human-readable format?

(a) Binary file
(b) Text file
(c) Object file
(d) Executable file

7. Binary files store data in

(a) ASCII format
(b) Machine format
(c) Character format
(d) Unicode text

8. Which operation is used to create a new file?

(a) Create
(b) Seek
(c) Read
(d) Delete

9. Which operation retrieves data from a file?

(a) Write
(b) Open
(c) Read
(d) Close

10. Which operation moves the file pointer to a specific position?

(a) Seek
(b) Read
(c) Write
(d) Delete

11. In sequential file organization, records are stored

(a) Randomly
(b) In key order
(c) In reverse order
(d) In encrypted form

12. Which file organization stores records in no particular order?

(a) Sequential
(b) Heap
(c) Indexed
(d) Tree

13. Which file organization provides fastest record access?

(a) Sequential
(b) Heap
(c) Hash
(d) Tape

14. An index contains

(a) Keys and pointers
(b) Only data
(c) Only keys
(d) Only pointers

15. The purpose of indexing is to

(a) Reduce file size
(b) Improve search speed
(c) Increase storage
(d) Encrypt data

16. A dense index contains entries for

(a) Every record
(b) Every block
(c) Every file
(d) Every database

17. Sparse index contains entries for

(a) Every record
(b) Some records or blocks
(c) Every bit
(d) Every byte

18. Primary index is created on

(a) Non-key field
(b) Primary key
(c) Random field
(d) File name

19. Secondary index is created on

(a) Primary key
(b) Non-key attribute
(c) File pointer
(d) Memory address

20. Which structure is commonly used in indexing?

(a) Stack
(b) Queue
(c) Tree
(d) Graph

21. B-tree is used in

(a) File indexing
(b) Sorting
(c) Encryption
(d) Networking

22. Overflow area is used in

(a) Sequential files
(b) Indexed sequential files
(c) Heap files
(d) Binary files

23. Which file organization is simplest to implement?

(a) Sequential
(b) Heap
(c) Hash
(d) Indexed

24. Which organization requires scanning entire file for searching?

(a) Sequential
(b) Indexed
(c) Hash
(d) Tree

25. File organization mainly affects

(a) Data retrieval time
(b) Screen resolution
(c) CPU speed
(d) Network bandwidth

26. In data hierarchy, a byte consists of

(a) 4 bits
(b) 8 bits
(c) 16 bits
(d) 32 bits

27. A field represents

(a) A single attribute
(b) A file
(c) A database
(d) A pointer

28. A record represents

(a) A collection of fields
(b) A collection of files
(c) A collection of databases
(d) A collection of bits

29. Which file type is easier for humans to read?

(a) Binary file
(b) Text file
(c) Object file
(d) Machine file

30. Which operation permanently removes a file?

(a) Delete
(b) Write
(c) Seek
(d) Read

31. Which organization uses a hash function?

(a) Sequential
(b) Hash
(c) Indexed
(d) Heap

32. Indexing reduces

(a) Storage
(b) Search time
(c) File size
(d) Record size

33. Which index stores entries for each record?

(a) Dense index
(b) Sparse index
(c) Block index
(d) File index

34. Which file organization uses key values to locate records directly?

(a) Sequential
(b) Hash
(c) Heap
(d) Tape

35. Which structure is used in modern database indexing?

(a) B-tree
(b) Stack
(c) Queue
(d) Linked list

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	2	B	3	A	4	B	5	D
6	B	7	B	8	A	9	C	10	A
11	B	12	B	13	C	14	A	15	B
16	A	17	B	18	B	19	B	20	C
21	A	22	B	23	B	24	A	25	A
26	B	27	A	28	A	29	B	30	A
31	B	32	B	33	A	34	B	35	A